

SAP API Specification Proposal

C&F Agent Interface · Shree Cement / Bangur

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Prepared by	SAP Backend Integration Team
Scope	10 S/4HANA APIs — 9 aligned to the client process register, 1 proposed addition
Distribution	Client SAP, Functional and Integration teams
Companion	CNF_API_Register.xlsx

DOCUMENT STATUS — BASIS OF PREPARATION

SAP system access (DEV / QAS / PRD)	Not yet granted
Backend release and component levels	Not confirmed
Repository inspection (SE80 / SE11 / SE37 / SEGW)	Not performed
Released-API and CDS availability check	Not performed
Customizing and master-data inspection	Not performed
Business sign-off on process rules	Not obtained

SAP object names, BAPI signatures, DDIC fields and released-API references in this document are **candidates**, derived from standard SAP release knowledge and the supplied functional narrative. The standard-versus-custom classification is an engineering estimate. **Section 9** lists the twenty-nine specific checks that convert each estimate into a verified fact.

This proposal establishes a shared technical vocabulary for the S/4HANA layer of the C&F Agent Interface, agrees the API inventory and its boundaries against the Commerce Cloud and Datasphere layers, separates what standard SAP already delivers from what must be built, and identifies precisely what must be confirmed in the client system before build commences. Its purpose is to make development access the immediate critical path.

1 Scope and architecture

The C&F Agent Interface presents depot and Clearing & Forwarding operations to the agent through a Spartacus storefront. Operational data for the product is distributed across **three systems** — SAP Commerce Cloud, SAP Datasphere and SAP S/4HANA — coordinated by SAP Integration Suite. This document specifies the **SAP S/4HANA-side** API surface only, and states explicitly where a concept is owned elsewhere.

In scope: S/4HANA read and command APIs, their candidate standard SAP sources, the custom development required where no standard equivalent exists, cross-cutting reliability and security requirements, and the validation programme.

Out of scope: storefront and frontend implementation, Commerce Cloud data modelling and Commerce-side APIs, integration flow construction, Datasphere model development, Basis configuration, functional process sign-off, and external statutory-service credential ownership.

1.1 Working topology



Assumption A-01 is confirmed. SAP Integration Suite sits in the runtime path between Commerce Cloud and S/4HANA. Earlier technical material showed a direct storefront-to-Gateway connection with no middleware; that reading is superseded. Transport-level retry, adapter configuration and correlation propagation are integration-layer responsibilities. What remains open is whether the integration layer or ABAP owns the outbound statutory calls and their credentials — **V-03**.

1.2 Source of record

S/4HANA is **not** the sole source of data for this product. It is authoritative for the business documents it creates and posts; substantial master and operational data is owned by Commerce Cloud, and analytical, consolidated and replicated data by Datasphere. Field-level mapping must respect this, and no API in this inventory should be assumed to be the authority for a value it merely displays.

Concept	Owning system	Currency	Note
Goods receipt, delivery, goods issue, shipment, shipment cost, billing document	S/4HANA	Real time	Created and posted in SAP. Authoritative document numbers
Stock and batch availability	S/4HANA	Real time	Transactional decision — must not be served from a replicated snapshot
IRN and E-Way Bill	Statutory portal , persisted in S/4HANA	Real time	Datasphere also carries an e-invoice object, source unconfirmed
Pending order list, deliveries and invoice billing as shown in the portal	Commerce T1	Pushed from SAP via CPI	The storefront reads these from Commerce over OCC, not from SAP directly
Invoice download	Commerce T1 , CRM portal fallback	—	SAP is not the primary retrieval path for the document file
Depot, user, agent-depot, geography master, material alias, Incoterms, storage location, depot-SL, SL-SPI	Commerce T2	Commerce-maintained	Portal master data. Not SAP-sourced — mapping to SAP organisational values must be defined
Vehicle master, transporter	Commerce T2 and Datasphere	Daily refresh in DSP	Duplicated. Ownership must be resolved — V-26
STO, MRN	Datasphere	Near real time	MRN is not generated by SAP in this architecture — see API-02
plant-SL, ageing report	Datasphere	Daily refresh	Analytical and reference reads

Three consequences for the SAP scope.

- 1. MRN is a Datasphere object, not a SAP output.** Earlier technical material derived the receipt note reference inside the goods-receipt function by prefixing the material document number. That is incorrect under this architecture: SAP should return the material document key and let MRN resolve from its owning source. See API-02 and V-09.
- 2. API-01 is narrower than a general pending-order read.** The portal's pending order list is served from Commerce T1. API-01 covers the SAP-side receipt position — what is awaiting goods receipt and what has been received — which is a different question from what the dashboard displays.
- 3. Storage location, depot-SL and SL-SPI originate in Commerce T2.** They are inputs to the SAP APIs, not values SAP derives. The mapping from Commerce depot and storage-location codes to SAP plant and storage-location values must be defined and owned — V-27.

Assumption A-02. For the documents it creates, S/4HANA is authoritative for business validity, posting, locking and document numbering. The storefront and Commerce layer render and initiate; they do not determine SAP business validity. This does not extend to concepts owned by Commerce or Datasphere, where SAP is a consumer or has no involvement.

2 Classification framework

Each API is classified by how much of its behaviour standard SAP already provides. This is the central question the document answers, and it drives both effort and risk.

Class	Meaning	ABAP work involved
S Standard	A delivered SAP API, BAPI or released service performs the business operation.	Thin service wrapper: protocol mapping, authorization, idempotency, error shaping. No business logic.
C Custom over standard	No single standard API returns the required shape, but every underlying source is standard.	Composite CDS view or application class over standard reads. Logic limited to derivation and aggregation.
X New construction	The required behaviour has no standard equivalent.	Designed and built: orchestration, state management, pre-document simulation, external coordination.

A class **S** rating does not mean no development is required — it means the *business operation* is delivered and the remaining work is integration scaffolding. A class **X** rating is where schedule and defect risk concentrate.

3 API inventory

Ten APIs. API-01 to API-09 correspond one-to-one with the client process register. API-10 is a proposed addition, discussed in §3.1. All ten are S/4HANA-side; Commerce-owned and Datasphere-owned reads are outside this inventory by design.

ID	API	Operation	Class	Primary candidate standard source	Service
API-01	Check MIGO	Read	C	Order, delivery and material document reads plus document flow	ZCNF_RECEIPT_SRV
API-02	Submit MIGO	Command	S	Goods movement posting	ZCNF_RECEIPT_SRV
API-03	Create DI	Command	S	Outbound delivery creation	ZCNF_DELIVERY_SRV
API-04	Stock Availability	Read	C	Stock and batch reads plus availability check	ZCNF_STOCK_SRV
API-05	Shipment Cost	Read + Estimate	X / S	Shipment costing (actual). No standard pre-document estimate	ZCNF_DISPATCH_SRV
API-06	Invoice Creation	Orchestration	X	Standard components per stage. No standard orchestration	ZCNF_DISPATCH_SRV
API-07	Invoice Correction	Command	S	Invoice correction request / billing cancellation	ZCNF_BILLING_SRV
API-08	E-Invoice Correction	Command	S	Document Compliance — IRN cancel and reissue	ZCNF_EDOC_SRV
API-09	E-Way Bill Extension	Command	S / X	Document Compliance for India	ZCNF_EDOC_SRV
API-10	Modify DI (<i>proposed</i>)	Command	S	Outbound delivery change	ZCNF_DELIVERY_SRV

Distribution: 5 Standard · 2 Custom over standard · 2 New construction · 2 conditional on Document Compliance licensing (API-08, API-09).

3.1 Three observations on inventory scope

E-Invoice Correction is correctly a separate API. The client register distinguishes Invoice Correction from E-Invoice Correction, and that distinction is right. Correcting the SAP billing document and correcting the statutory e-invoice are different operations with different constraints: a generated IRN cannot be amended at all, only cancelled within a statutory window and reissued. Treating them as one API would force incompatible rules into a single code path. They are specified separately as API-07 and API-08.

Modify DI is absent from the client register and is proposed here. Create DI has no amend path without it, and a delivery must be correctable before goods issue — quantity revision, batch reallocation and vehicle substitution are all routine before dispatch. It is classified Standard and carries the smallest effort in the inventory. Confirmation of scope is **V-25**.

Several portal capabilities are deliberately not SAP APIs. Pending order display, deliveries and invoice-billing lists, invoice download, and the depot, user, geography, material-alias, transporter, vehicle and Incoterm masters are served from Commerce Cloud or Datasphere. They appear in this document only where they are *inputs* to a SAP API, and their ownership is recorded in §1.2 rather than claimed here.

3.2 Service decomposition

A single OData project for ten APIs creates a build and transport bottleneck for a two-developer team, and one faulty activation affects every endpoint. The inventory is therefore grouped into five services by document domain, each independently transportable and independently ownable.

Service	APIs	Domain
ZCNF_RECEIPT_SRV	API-01, API-02	Inbound receipt and goods movement
ZCNF_DELIVERY_SRV	API-03, API-10	Delivery lifecycle
ZCNF_STOCK_SRV	API-04	Stock and batch availability
ZCNF_DISPATCH_SRV	API-05, API-06	Freight, dispatch and billing execution
ZCNF_BILLING_SRV / ZCNF_EDOC_SRV	API-07 / API-08, API-09	Billing correction and statutory documents

4 Functional specifications

API-01 Check MIGO C

Purpose. Return the SAP-side receipt position — inbound consignments awaiting goods receipt and those already received — so pending quantities are visible and receipt can be initiated.

This is not the portal's pending-order dashboard, which is served from Commerce T1 over OCC (§1.2). API-01 answers a narrower question: what is awaiting goods receipt in SAP, and what has already been received against it.

Input	Requirement	Notes
Plant	Mandatory	Bounds the result set and enforces authorization
Document number	Optional	Single-document lookup
Date range from / to	Mandatory	Bounds the query — see NFR-04
Receipt status filter	Optional	Pending / Partial / Complete

Outputs. Per line: source document and item, material, description, brand and grade, plant, ordered quantity, received quantity, **pending quantity**, unit of measure, document dates, receipt status, and material document reference. The receipt note (MRN) is a Datasphere-owned object and is **not** derived here — SAP returns the material document key against which MRN resolves in its owning system.

BUSINESS RULES

- **BR-01.1** Receipt status must distinguish **Pending / Partial / Complete**. A binary received-or-not indicator is insufficient: partial truck receipts are normal in depot replenishment.
- **BR-01.2** Pending quantity = ordered quantity – cumulative received quantity, derived from document flow, not stored.
- **BR-01.3** Results restricted to plants the user is authorized for.

API-02 Submit MIGO S

Purpose. Post the goods receipt for a received consignment, creating the SAP material document and returning the receipt reference the portal displays.

Inputs. Reference document and item; posting date and document date; external reference (challan); line items carrying material, plant, storage location, quantity, unit of measure and batch; external request identifier.

Outputs. Material document number and year, per-item status, correlation identifier.

BUSINESS RULES

- **BR-02.1** Receipt quantity must not exceed pending quantity.
- **BR-02.2** Partial receipt permitted; residual remains pending.
- **BR-02.3** Posting is rejected in full if any line fails. No partial commit.
- **BR-02.4** A repeated call carrying an already-processed request identifier returns the original result and does **not** post again.

MRN is not a SAP output. Earlier technical material derived the receipt note reference inside the goods-receipt function by prefixing the material document number — a developer convention, not a business identifier. Under the confirmed architecture MRN is a Datasphere object (§1.2), so this API must not invent it. SAP returns the material

document key; MRN resolves in its owning system. The linkage rule and whether the portal needs MRN echoed back through SAP at all is **V-09**.

API-03 Create DI S

Purpose. Create the outbound delivery that authorizes dispatch against a source demand document, capturing the logistics detail the depot records at that point. DI is the outbound delivery; the DI number is the delivery number.

Inputs. Source document and item; shipping point; planned goods-issue and delivery dates; storage location at header level; line items with material, quantity, unit of measure and batch allocation; vehicle, driver name, driver mobile and transporter; external request identifier.

Storage location, depot-SL and SL-SPI originate in Commerce T2, and transporter and vehicle master are held in both Commerce T2 and Datasphere (§1.2). They are *inputs* to this API, not values SAP derives. The mapping from Commerce depot and storage-location codes to SAP plant and storage-location values must be defined and owned — **V-27** — and the duplicated transporter and vehicle sources reconciled — **V-26**.

BUSINESS RULES

- **BR-03.1** Delivery quantity must not exceed open source quantity.
- **BR-03.2** Exactly **one storage location** applies per delivery, held at header level. Multiple batches may be allocated within it.
- **BR-03.3** Allocated batch quantities must sum **exactly** to the delivery quantity. Neither under- nor over-allocation is permitted.
- **BR-03.4** Driver mobile is mandatory where dispatch documents are issued by SMS.
- **BR-03.5** Creation is atomic: the delivery and its logistics attributes commit together or not at all.

API-04 Stock Availability

Purpose. Present available stock at plant, storage-location and batch grain so the depot user can select a storage location and allocate batches when preparing a dispatch. This API is the precondition for BR-03.2 and BR-03.3.

Input	Requirement	Notes
Plant	Mandatory	Enforces authorization and bounds the query
Material	Mandatory	Single material or restricted list
Storage location	Optional	Omit to return all locations for the plant
Batch	Optional	Single-batch lookup
Stock semantic	Mandatory	Unrestricted or availability-check — see V-07
Allocatable only	Optional	Suppress blocked and quality-inspection stock

Outputs. Per row: plant, storage location, material, description, brand and grade, batch, unrestricted quantity, blocked quantity, quality-inspection quantity, availability quantity, unit of measure, manufacture date, shelf-life expiry date, ageing in days, allocation rank, allocatable indicator.

BUSINESS RULES

- **BR-04.1** Only storage locations belonging to the requested plant are returned.
- **BR-04.2** Only unrestricted stock is allocatable. Blocked and quality-inspection stock is displayed for visibility but flagged not allocatable.
- **BR-04.3** Allocation rank is derived by ascending manufacture date where batch age governs issue sequence. Whether SAP determines this sequence or the portal proposes it is unresolved — **V-08**.
- **BR-04.4** Where shelf life applies, batches past expiry are excluded from the allocatable set and reported separately.
- **BR-04.5** Quantities returned in base unit of measure with the unit stated explicitly. No silent conversion.
- **BR-04.6** The response is a point-in-time read. It carries no reservation and confers no guarantee that stock remains available at posting.

Non-functional. Mandatory plant and material filters, server-side pagination with a stated maximum page size, deterministic ordering. Called interactively during data entry — response target 2 seconds at the 95th percentile.

API-05 Shipment Cost

Purpose. Two related but architecturally distinct needs, specified as separate modes.

Mode ESTIMATE — before any document exists, return an indicative freight cost for a proposed dispatch so the user can validate it against expectation.

Mode ACTUAL — after shipment costing has been posted, return the calculated cost and its accounting position.

MODE ESTIMATE INPUTS

Source plant, destination, material, quantity and unit of measure, Incoterm (all mandatory); transporter, vehicle capacity class and route (optional, derived where omitted).

Outputs. Estimated cost, currency, rate basis (per tonne / per trip / per kilometre), distance, resolved route, condition-level breakdown, rate validity, and an explicit indicator of whether a rate was found.

BUSINESS RULES

- **BR-05.1** The estimate is non-binding and **creates no SAP document**. No shipment, no cost document, no number-range consumption.
- **BR-05.2** Where no freight rate is maintained for the route and transporter combination, the API returns an explicit no-rate condition. **It must never return zero cost**, which the portal would render as free carriage.
- **BR-05.3** Incoterm governs whether the estimate is chargeable to the customer or absorbed. The two cases must be distinguishable in the response.
- **BR-05.4** Estimated and actual cost must be separately identifiable throughout. An estimate is never presented as a posted cost.
- **BR-05.5** Mode ACTUAL is read-only. Cost correction is a separate process.

Why this is new construction. Standard SAP shipment costing is *document-bound*: freight is calculated on a shipment cost document, which requires a shipment, which requires deliveries. There is no standard mechanism for a pre-commitment estimate. Mode ESTIMATE is therefore genuinely new construction; Mode ACTUAL is a standard read. Options are compared in \$5.

API-06 Invoice Creation X

Purpose. Execute the dispatch-to-invoice sequence for a prepared delivery — post goods issue, create the shipment, calculate and post shipment cost, create the billing document, and initiate the statutory e-invoice and E-Way Bill — returning every resulting document identifier and the status of each stage. This is the portal's "Generate Invoice" action and the highest-value operation in the inventory.

Inputs. Delivery number; shipment detail (vehicle, driver name, driver mobile, consignment note number and date); transporter; confirmation of batch allocation; external request identifier; optional stage selection.

STAGE MODEL

1 Goods issue	-> material document
2 Shipment	-> shipment document
3 Shipment cost	-> shipment cost document
4 Billing	-> billing document + accounting document
5 Statutory e-invoice	-> invoice reference number (IRN)
6 E-Way Bill	-> E-Way Bill number + validity

Outputs. Process identifier, overall state, and per stage: name, state (Not started / In progress / Complete / Failed / Skipped), resulting document key, timestamp, and error detail where failed.

BUSINESS RULES

- **BR-06.1** Stages execute in sequence. Each stage's precondition is the successful completion of its predecessor.
- **BR-06.2 The sequence is not a single unit of work.** Goods issue, shipment, shipment cost and billing are separate SAP commits; stages 5 and 6 are external calls with unpredictable latency and independent failure. A failure at stage n does not and must not roll back stages 1 to $n-1$, which are already committed and, in the case of goods issue, have moved physical stock.
- **BR-06.3** The process must be **resumable**. A failed stage is retried from that stage, not from the beginning. Retrying must never repeat a completed stage.
- **BR-06.4** Partial completion is reported explicitly, never concealed. The portal must be able to render "goods issue posted, billing failed" as a distinct and actionable state.
- **BR-06.5** Idempotent at both levels: a repeated call with a known request identifier returns current state without re-executing, and each stage is individually guarded.
- **BR-06.6** Every stage records its correlation identifier and resulting document key.
- **BR-06.7** Synchronous completion versus immediate return with a process identifier is an architectural decision — **V-04**. This specification assumes immediate return, because BR-06.2 makes reliable synchronous completion unachievable across six stages including two external calls.

Non-functional. Concurrency control per delivery — two simultaneous invocations for the same delivery must not both proceed. Stage-level timeout. Reconciliation endpoint for the "SAP committed, response lost" case.

API-07 Invoice Correction S

Purpose. Correct a billed dispatch where quantity or value is wrong, or reverse it entirely.

Inputs. Original billing document; correction type (correct or reverse); reason code; item-level quantity and value corrections where applicable; external request identifier.

BUSINESS RULES

- **BR-07.1** Correction and reversal are distinct SAP processes and must not share one code path.

- **BR-07.2** Corrections are made by a new correcting document referencing the original. The original is never altered.
- **BR-07.3** Where the original carries a statutory invoice reference, the correction must first evaluate the e-invoice position through API-08 before any billing action is taken.

API-08 E-Invoice Correction S

Purpose. Correct the statutory e-invoice for a billed dispatch. This is a separate operation from API-07 because a generated IRN cannot be amended — correction is cancel-and-reissue inside the statutory window, and a credit or debit note outside it.

Inputs. Billing document or IRN; correction reason; corrected values where a reissue is required; external request identifier.

Outputs. Action taken (cancelled and reissued, or routed to credit/debit note), new IRN where reissued, statutory response reference, E-Way Bill position, window status with time remaining.

BUSINESS RULES

- **BR-08.1** A generated IRN cannot be amended. Correction is cancel-and-reissue within the statutory window only.
- **BR-08.2** Inside the statutory cancellation window the API cancels the IRN and reissues. Outside it, the API must refuse cancellation and route the correction to a credit or debit note instead.
- **BR-08.3** Cancellation is blocked where an active E-Way Bill is linked to the invoice. The E-Way Bill position must be evaluated first.
- **BR-08.4** The statutory window is evaluated server-side from the IRN generation timestamp, never from a client-supplied value.
- **BR-08.5** Every cancellation and reissue attempt is recorded with its statutory response for audit, successful or failed.
- **BR-08.6** The exact window duration and its edge behaviour must be confirmed against current regulation and the installed component — **V-14**.

API-09 E-Way Bill Extension S X

Purpose. Extend the validity of an active E-Way Bill where the consignment cannot complete within the original window.

Inputs. E-Way Bill number; current vehicle number; current location and state code; extension reason code and remarks; external request identifier.

BUSINESS RULES

- **BR-09.1** Transport mode is **Road only**. No mode parameter is exposed.
- **BR-09.2** Extension period is 24 hours and may be repeated.
- **BR-09.3** Vehicle change is a Part B update, not an extension. The two must not be conflated.
- **BR-09.4** Every attempt, successful or failed, is recorded with its statutory response for audit.
- **BR-09.5** Reason codes must match the current statutory code list — **V-13**.

API-10 Modify DI S *proposed addition*

Purpose. Amend an open delivery before goods issue.

Inputs. Delivery number; revised quantity; revised logistics attributes within the agreed scope; external request identifier.

BUSINESS RULES

- **BR-10.1** Modification rejected once goods issue is posted.
- **BR-10.2** Editable field scope requires confirmation — **V-11**.
- **BR-10.3** Revised quantity remains subject to BR-03.1 and BR-03.3.

5 Technical design

5.1 Common layering pattern

```
OData V2 request
-> DPC_EXT method          protocol mapping only, no business logic
-> request / response mapper
-> application service class business orchestration, testable in isolation
-> validator                business preconditions
-> API / BAPI wrapper       isolates the standard SAP call
-> standard SAP API or query
-> SAP business document
```

Generated Gateway classes remain thin. Business logic resides in classes rather than function modules, so ABAP Unit tests with test doubles are possible. Each standard SAP call is reached through a wrapper, so a change of underlying API — for example following V-01 — affects one class rather than the service contract.

5.2 Candidate standard sources

API	Candidate standard source	Must be confirmed
API-01	Order, delivery and material document reads; document flow. Released CDS views where the release provides them	Whether released views exist; build composite over them rather than raw table selects
API-02	Goods movement creation BAPI; released material document service on newer releases	Movement type and reference model; whether the standard service can be called directly
API-03	Outbound delivery creation — sales-order-referenced or STO-referenced variant	V-05. These are different BAPIs and the wrong one will not run
API-04	Availability check BAPI; stock and batch reads; batch master detail; released stock service on newer releases	V-07 stock semantic; V-21 batch management active
API-05	Shipment cost creation and detail retrieval (classic transport). No standard estimate mechanism	V-06 transport solution; V-10 simulation feasibility
API-06	Per stage: delivery goods-issue confirmation, shipment creation, shipment cost creation, billing creation, Document Compliance for statutory stages	V-04, V-06, V-02. One classic delivery-update module in common use is not released and should not be called from a custom service
API-07	Invoice correction request document type; billing cancellation BAPI	Whether the correction request type is configured; V-22 identifier requirements
API-08	Document Compliance for India — IRN cancellation, status and reissue	V-02 licensing; V-14 statutory window
API-09	Document Compliance for India — E-Way Bill actions	V-02 licensing; V-13 reason codes
API-10	Outbound delivery change BAPI — the outbound structure family, not the inbound one	V-11 editable scope

5.3 API-05 — three options for the pre-document estimate

Option	Approach	Assessment
1	Read freight condition records directly and apply rate logic in custom ABAP	Workable, but replicates pricing logic and will diverge from actual costing as condition configuration evolves. Divergence is silent and erodes user trust in the estimate
2	Invoke standard condition-technique pricing in simulation, creating no document	Recommended. Reuses the client's actual freight condition configuration, so estimate and actual converge by construction. Requires validation that no document or number range is consumed — V-10
3	Create shipment and cost document, read the result, reverse	Rejected. Consumes number ranges, writes and reverses real documents, pollutes document flow and audit trail. Unacceptable for an interactive estimate

5.4 API-06 — required custom components

Every individual stage is a standard SAP operation. What has no standard equivalent is the orchestration: a resumable, observable, idempotent, multi-commit process spanning four SAP units of work and two external calls.

Component	Purpose
Process state persistence	Process identifier, delivery, stage, state, document keys, timestamps, error detail
Stage engine	Precondition evaluation, sequencing, resume from failed stage
Per-stage guard	Prevents re-execution of a completed stage
Process-level idempotency	Request identifier to process identifier association
Concurrency control	Prevents two simultaneous processes for one delivery
Reconciliation query	Resolves the "SAP committed, response lost" case

6 Cross-cutting requirements

These apply to every API and are built once as shared components, before any command endpoint exists.

Ref	Requirement	Standard available?
NFR-01	Idempotency. Every command accepts an external request identifier. A repeat of a processed request returns the original result without re-executing. The association must survive the "committed but response lost" case	No. Custom by necessity. The most commonly underestimated item in the inventory
NFR-02	Correlation and logging. Every call records its correlation identifier against the resulting document, queryable for reconciliation	Yes — standard application log framework. Should not be rebuilt as custom tables
NFR-03	Error contract. Stable structure: HTTP status, business error code, SAP message reference, safe message text, correlation identifier, retryable flag, field context. Business validation must be distinguishable from schema failure, conflict and temporary condition	Partly — the Gateway exception framework provides the mechanism
NFR-04	Bounded reads. Mandatory filters, server-side pagination, stated maximum page size, deterministic ordering	Standard OData facility
NFR-05	Authorization. Plant and depot scoping enforced on every operation. Custom authorization objects supplement — never replace — standard SAP document authority, which executes inside the standard APIs regardless	Yes — standard objects exist for goods movement, delivery and billing

Ref	Requirement	Standard available?
NFR-06	Type discipline. Leading zeros preserved on SAP keys; decimal precision paired with unit of measure; explicit date and timezone rules; no internal type semantics crossing the boundary	Convention
NFR-07	No credentials in application tables. External statutory-service credentials held in the platform credential store or SAP secure storage	Yes

7 Standard versus custom analysis

ID	API	Class	What standard provides	What must be custom	Risk
API-01	Check MIGO	C	All underlying document reads	Cross-document composite, tri-state status, pending-quantity derivation	Med
API-02	Submit MIGO	S	The goods-receipt posting in full	Idempotency, receipt-note derivation, plant authorization, error shaping	Low
API-03	Create DI	S	Delivery creation in full	Logistics attribute persistence, source-document strategy	Med
API-04	Stock Availability	C	Stock reads, availability check, batch master	Composite view, allocation ranking, allocatable determination, expiry exclusion	Med
API-05	Shipment Cost	X / S	Posted cost retrieval (ACTUAL)	The entire pre-document estimate	High
API-06	Invoice Creation	X	Every individual stage operation	Orchestration, state model, resumption, process idempotency	High
API-07	Invoice Correction	S	Correction request document and cancellation	Statutory window guard, routing between two processes	Med
API-08	E-Invoice Correction	S	IRN cancellation, status and reissue via Document Compliance	Window evaluation and branch logic; E-Way linkage guard	Med
API-09	E-Way Bill Extension	S / X	E-document lifecycle and statutory connectivity, if licensed	Near nothing if licensed; a full statutory client if not	High
API-10	Modify DI	S	Delivery change in full	Editable-scope enforcement	Low

Where the risk sits. API-05 ESTIMATE and API-06 account for the substantial majority of genuine construction and schedule risk. Five APIs are standard operations requiring integration scaffolding only.

The licensing swing. API-08 and API-09 both depend on SAP Document Compliance for India. If it is licensed and configured, both are near-standard. If it is not, API-09 alone grows by 12–15 days and API-08 requires a bespoke statutory client with its own status store. This is **one question (V-02)** with the largest single effect on the estimate — and it is answerable in minutes with system access.

8 Indicative effort

Pre-access ranges, stated to support sequencing rather than to fix a schedule. These narrow once §9 is closed.

Item	Class	Low	High	Note
Cross-cutting foundations (§6)	—	15	20	No standard equivalent for idempotency
API-01 Check MIGO	C	8	12	Composite read
API-02 Submit MIGO	S	5	8	Standard posting plus scaffolding
API-03 Create DI	S	6	10	Plus logistics attributes

Item	Class	Low	High	Note
API-04 Stock Availability	C	8	12	Composite read with ranking
API-05 Shipment Cost	X / S	12	18	Estimate is new construction
API-06 Invoice Creation	X	25	35	Highest risk item
API-07 Invoice Correction	S	5	8	Plus statutory guard
API-08 E-Invoice Correction	S	6	10	Assumes Document Compliance
API-09 E-Way Bill Extension	S	3	5	Assumes Document Compliance
API-10 Modify DI	S	4	6	Proposed addition
Indicative total	—	97	144	Document Compliance available
Contingency if Document Compliance absent	X	+20	+25	API-08 and API-09 custom route

Excludes functional analysis, integration-layer work, testing cycles and defect resolution.

9 Validation register

Twenty-nine items. Each is a specific check that converts an estimate in this document into a verified fact, ranked by how much of the inventory it gates. **This register is the substance of the development-access request.**

Ref	Validation required	Gates	Owner
V-01	Backend product and exact release / component levels. Which released APIs and CDS views are available and permitted	All ten — determines whether several APIs need custom construction at all	Basis / ABAP
V-02	Is SAP Document Compliance for India licensed, installed and configured? Which statutory actions are delivered — IRN generate, IRN cancel, E-Way Part B, E-Way extension?	API-08, API-09, statutory stages of API-06. Largest single effect on the estimate	Basis / Tax
V-03	Integration Suite is confirmed in the runtime path. Does it own the outbound statutory calls and their credentials, or is ABAP expected to call the statutory service directly?	API-08, API-09, NFR-07	Architect / CPI
V-04	Is the dispatch-to-invoice sequence a single synchronous command or a staged process?	API-06 entirely, API-05	SD / Architect
V-05	Is the source demand document for a delivery a sales order or a stock transport order? Does this differ by business scenario?	API-03, API-10, API-01	SD
V-06	Which transport solution is in use — classic shipment and shipment costing, or an alternative?	API-05 both modes, shipment stages of API-06	SD / LE
V-07	What does "available stock" mean operationally — unrestricted physical, availability-check result, or batch-eligible?	API-04, API-03	MM / Business
V-08	Is batch issue sequence determined by SAP configuration or proposed by the portal?	API-04, API-03	MM / SD
V-09	MRN is a Datasphere object. What links it to the SAP material document, and does the portal require it echoed through SAP at all?	API-01, API-02	MM / Datasphere
V-10	Do route, freight condition and rate configuration support pricing simulation without document creation?	API-05 ESTIMATE mode	SD / LE
V-11	Which delivery fields are editable after creation, and until which status?	API-10	SD
V-12	Master and reference code dictionaries: brand and grade derivation, material grouping, business segment, organisational codes, Incoterm treatment	Every payload field across all ten	SD / Master data
V-13	Current statutory reason-code list for E-Way Bill extension	API-09	Tax
V-14	Statutory cancellation window for e-invoice, its edge behaviour, and E-Way Bill linkage rules on cancellation	API-08, API-07, API-06	Tax / SD
V-15	Landscape: development, quality and production clients, and the transport route	All — a two-client route with no quality client is not appropriate for APIs with financial impact	Basis

Ref	Validation required	Gates	Owner
V-16	Gateway topology: embedded or hub. Service registration and activation process	All	Basis
V-17	Package, namespace, message class, exception and transport conventions	All — must be agreed before either developer writes a class	ABAP lead
V-18	Approved error schema and HTTP status mapping across the landscape	NFR-03, all	Architect
V-19	Correlation and logging standard end to end	NFR-02, all	Architect
V-20	Where the SAP document persists an external request identifier, per document type	NFR-01, all commands	ABAP
V-21	Is batch management active for all relevant materials? Does shelf life apply?	API-04, API-03	MM
V-22	Which invoice identifier the portal requires at each point — billing document, IRN, accounting document, or display number	API-07, API-08, API-06	SD / FI
V-23	Standard authorization objects in force, and whether existing depot roles already scope by plant	NFR-05, all	Basis / Security
V-24	Where statutory and invoice output documents are stored and how they are retrieved	Portal download requirement	SD / Tax
V-25	Is Modify DI in scope? Absent from the client register, but Create DI has no amend path without it	API-10	Client / SD
V-26	Transporter and vehicle master exist in both Commerce T2 and Datasphere. Which is authoritative, and which value does SAP receive at posting?	API-03, API-05, API-06, API-10	Architect / Commerce / Datasphere
V-27	Mapping from Commerce depot, storage location, depot-SL and SL-SPI codes to SAP plant and storage-location values. Who owns the mapping and where does it execute — Commerce, CPI or SAP?	API-01 to API-06, API-10	Architect / SD / Commerce
V-28	Datasphere carries an e-invoice object of unconfirmed source. Does it duplicate the SAP e-document status, and which does the portal read?	API-08, API-06	Datasphere / Tax
V-29	What does SAP push to Commerce T1 for the pending-order, deliveries and invoice-billing views, on what trigger and at what latency? Is that an existing flow or new scope?	Boundary of API-01; portal dashboard	Architect / CPI / Commerce

10 Proposed next steps

1. **Grant development access.** V-01, V-02, V-06, V-15, V-16, V-21 and V-23 are answerable within the first two days and together resolve the classification of six of the ten APIs.
2. **Resolve the cross-system source-of-record items** — V-03 (statutory call and credential ownership), V-26 (duplicated transporter and vehicle master), V-27 (Commerce-to-SAP organisational mapping), V-28 (duplicated e-invoice status) and V-29 (what SAP pushes to Commerce T1). These are architecture decisions across three systems and none of them is an ABAP decision to make alone.

3. **Hold a functional session** on V-04, V-05, V-07, V-08, V-09, V-11, V-22 and V-25 with the responsible process owners.
4. **Deliver master-data knowledge transfer** against V-12. This gates field-level correctness on every API and cannot be substituted by system inspection alone.
5. **Agree development conventions** (V-17) before build starts, with both ABAP developers present.
6. **Begin build with API-04 and API-01** — both read-only, both create no business state, and together they prove source selection, filtering, pagination, authorization, the error contract, correlation and connectivity with zero posting risk. Commands follow once the pattern is proven.

On the basis of items 1 to 3 being closed, the validation register can be substantially retired within two weeks of access, at which point this proposal converts to a verified specification and the effort ranges in §8 narrow.

Appendix A Assumption log

Ref	Assumption	Impact if wrong	Confidence
A-01	SAP Integration Suite sits between Commerce Cloud and S/4HANA and owns transport retry	—	Confirmed by client architecture
A-02	For the documents it creates, S/4HANA is authoritative for business validity, posting, locking and numbering. It is not the source of record for Commerce- or Datasphere-owned concepts	Validation would need to move layer	High
A-10	Commerce T2 masters (depot, storage location, depot-SL, SL-SPI, Incoterms, material alias) are inputs to the SAP APIs, not SAP-derived values	Mapping ownership and derivation move into SAP scope	Confirmed by client architecture — mapping owner still open, V-27
A-11	MRN, STO, plant-SL and ageing are Datasphere-owned; SAP does not generate MRN	API-02 would need an MRN derivation rule and an owner	Confirmed by client architecture — V-09
A-03	DI is the outbound delivery; DI number is the delivery number	Delivery-related APIs change object	Confirmed by client team
A-04	One storage location per dispatch, multiple batches within it	Allocation model and API-04 response shape change	Medium
A-05	The dispatch-to-invoice sequence spans multiple SAP units of work	If atomic, API-06 redesigns entirely	High on engineering grounds; needs functional confirmation
A-06	Classic shipment and shipment costing are in use	API-05 and the shipment stages of API-06 redesign	Unverified — V-06
A-07	SAP Document Compliance for India is licensed and handles IRN and E-Way Bill	API-08 and API-09 both expand substantially	Unverified — V-02
A-08	Batch management is active for dispatched materials	API-04 response shape simplifies	Medium
A-09	Modify DI is required even though absent from the client register	If out of scope, API-10 is removed and Create DI has no amend path	Proposed — V-25

Appendix B Document control

Version	Date	Change	Status
1.0	29 Jul 2026	Initial proposal. Nine APIs specified, classification framework and validation register established	Pre-access proposal
1.1	29 Jul 2026	Realigned to the client process register. E-Invoice Correction separated from Invoice Correction as API-08. Modify DI retained as a proposed addition (API-10). Service decomposition and effort updated	Pre-access proposal
1.2	29 Jul 2026	Architecture revised against the client landscape diagram. Integration Suite confirmed in the runtime path (A-01 closed). Added the cross-system source-of-record model (§1.2) recognising Commerce Cloud and Datasphere as owners of substantial master and operational data. MRN reclassified as Datasphere-owned and removed from API-02 output. API-01 scope narrowed against the Commerce-served dashboard. Added V-26 to V-29 and A-10, A-11	Pre-access proposal

Review requested from: client SAP technical lead, SD and MM functional owners, integration architect, Basis.

Response requested on: the validation register (§9), the architecture assumption (A-01), the API-06 synchronicity decision (V-04), the cross-system source-of-record items (V-26 to V-29), and the scope of Modify DI (V-25).